CHAPTER 6

The logarithm function

S44, S47, S49, S50

> **Key point · Objectives.** Use the properties of ln; solve equations and thresholds; differentiate and study logarithmic functions.

Course notes

Definition and properties

The exponential is continuous and strictly increasing on $\mathbb{R}$, and its range is $]0,+\infty[$. For every $x>0$, there is a unique real number y such that $e^y = x$: we denote it by $\ln x$.

$$\ln(e^y) = y \quad (y \in \mathbb{R}), \qquad e^{\ln x} = x \quad (x>0), \qquad \ln 1 = 0, \quad \ln e = 1.$$

For $a, b > 0$ and $n \in \mathbb{Z}$,

$$\ln(ab) = \ln a + \ln b, \quad \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \quad \ln(a^n) = n \ln a, \quad \ln\sqrt{a} = \tfrac{1}{2}\ln a.$$

Proof of the product rule. $e^{\ln a + \ln b} = e^{\ln a} e^{\ln b} = ab$; the uniqueness of the preimage gives the formula. There is no analogous formula for $\ln(a+b)$.

Derivative, variations and limits

We accept that ln is differentiable on $]0,+\infty[$. Differentiating $e^{\ln x} = x$ gives $e^{\ln x}(\ln x)' = 1$; hence $(\ln x)' = \frac{1}{x} > 0$. ln is strictly increasing and concave, because its second derivative is $-\frac{1}{x^2}$.

$$\lim_{x\to 0^+} \ln x = -\infty, \qquad \lim_{x\to+\infty} \ln x = +\infty.$$

Its curve is the reflection of that of the exponential in the line $y = x$. The line $x = 0$ is a vertical asymptote. Its tangent at 1 gives $\ln x \leq x - 1$ for $x > 0$.

> **Key point · Composition and comparison of growth rates.** If $u > 0$ is differentiable, $(\ln u)' = \frac{u'}{u}$. For every integer $m \geq 1$,
>
> $$\lim_{x\to+\infty} \frac{\ln x}{x^m} = 0, \qquad \lim_{x\to 0^+} x^m \ln x = 0.$$

Proof at 0. Set $t = -\ln x$; as $x \to 0^+$, $t \to +\infty$ and $x \ln x = -te^{-t} \to 0$ by comparison of growth rates. For $m \geq 1$, the same reasoning gives $-te^{-mt} \to 0$. At $+\infty$, set $t = \ln x$ to write $\frac{\ln x}{x^m} = \frac{t}{e^{mt}} \to 0$.

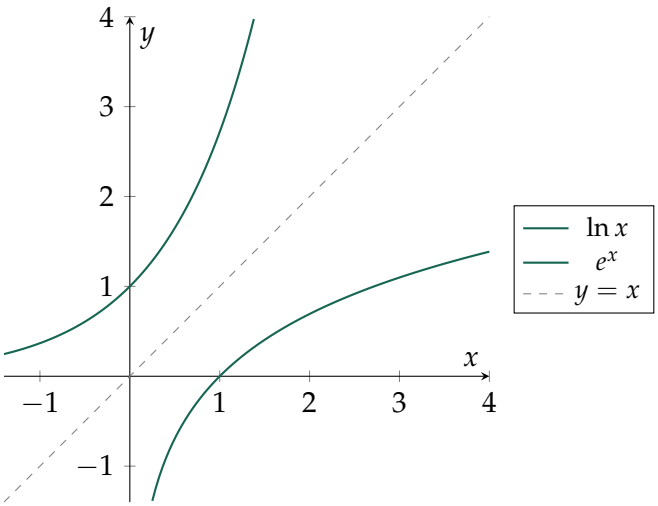
Equations and thresholds

For $A, B > 0$, $\ln A = \ln B \Leftrightarrow A = B$ and $\ln A \leq \ln B \Leftrightarrow A \leq B$. Always start with the domain.

Example. $\ln(x-1) + \ln(x+1) = \ln 8$ requires $x > 1$. The equation becomes $x^2 - 1 = 8$, so $x = 3$ only.

Threshold. $0.8^n \leq 0.1$ is equivalent to $n \ln 0.8 \leq \ln 0.1$, then $n \geq \frac{\ln 0.1}{\ln 0.8} \simeq 10.319$. The first integer is 11.

Example of a study. For $f(x) = \frac{\ln x}{x}$, $f' = \frac{1-\ln x}{x^2}$; f increases on $]0, e]$, decreases on $[e, +\infty[$, and its maximum is $\frac{1}{e}$. Its limit at 0^+ is $-\infty$, its limit at $+\infty$ is 0.



Research activity

Controlling a calculation error — 30 min.

A calculation method produces an initial error $e_0 = 5$. At each step, the error is multiplied by 0.7. Denote by e_n the error after n steps, for $n \in \mathbb{N}$.

1. Express e_n in terms of n.
2. Using the logarithm, determine the smallest integer n such that $e_n \leq 10^{-4}$.
3. Using a calculator, calculate the error at the value found and at the previous value to check that it is minimal.
4. We now require an error of at most 10^{-6}. Determine the new minimum number of steps. Does dividing the tolerance by 100 multiply the number of steps by 100?

Progressive exercises

E1 · Practice. After determining the set on which it is defined, solve the equation $\ln(2x-1) = \ln(x+3)$.

E2 · Practice. Determine the real numbers x for which $\ln x + \ln(x-2)$ is defined, then solve $\ln x + \ln(x-2) = \ln 3$.

E3 · Practice. Consider $f(x) = \ln(x^2+1)$ and $g(x) = x\ln x$. State their domains of definition and calculate their derivatives.

E4 · Practice. Determine $\lim\limits_{x\to+\infty} \dfrac{\ln x}{\sqrt{x}}$ and $\lim\limits_{x\to 0^+} x^2 \ln x$. Justify the comparison-of-growth-rates results used.

E5 · Practice. By studying a function, prove that $\ln x \leq x-1$ for every real number $x > 0$. Do not use the concavity of the logarithm directly.

Textbook or past-paper work

Barbazo, p. 196, exercise 36. At home after S44: do the three functions. Write down the conditions of definition before solving the inequalities; justify the open endpoints of the intervals obtained.

Homework

Homework 6 — Study of a logarithmic function — 55 min.

Consider the function f defined on $(0, +\infty)$ by $f(x) = \frac{1+\ln x}{x}$. Its graph is denoted $\mathcal{C}_f$.

1. Determine the limits of f at the endpoints of its domain, study its sign, then draw up its table of variations.
2. Study the convexity of f and determine the coordinates of the point of inflection of $\mathcal{C}_f$.
3. Prove that the equation $f(x) = \frac{1}{2}$ has exactly two solutions. Give for each a justified bracketing using calculated values.
4. Using the maximum of f, prove that $1 + \ln x \leq x$ for all $x > 0$. State the case of equality.